

Home work 1 (5.4)

#5 : Length of $y = x^{2/3}$ from $x=1$ to $x=8$.

Formulas: (1) $L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$

OR (2) $L = \int_c^d \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$

we have $y = x^{2/3}$

$$\Rightarrow y^{3/2} = \left(x^{2/3}\right)^{3/2}$$

$$\Rightarrow y^{3/2} = x^{2/3 \cdot 3/2}$$

$$\Rightarrow y^{3/2} = x$$

we will use the second formula:

Limits for y :

x	1	8
y	1	$8^{2/3} = 4$

$$\frac{dx}{dy} = \left(y^{3/2}\right)' = \frac{3}{2} y^{3/2-1} = \frac{3}{2} y^{1/2}$$

$$\Rightarrow L = \int_1^4 \sqrt{1 + \left(\frac{3}{2} y^{1/2}\right)^2} dy$$

$$L = \int_1^4 \sqrt{1 + \frac{9}{4}y} \, dy$$

Substitution: $u = 1 + \frac{9}{4}y$

$$\Rightarrow du = \frac{9}{4} dy$$

$$\Rightarrow dy = \frac{4}{9} du$$

y	1	4
u	$\frac{13}{4}$	10

$$\Rightarrow L = \int_{13/4}^{10} \sqrt{u} \cdot \frac{4}{9} du$$

$$L = \int_{13/4}^{10} u^{1/2} \cdot \frac{4}{9} du$$

note:

$$\int u^n du = \frac{u^{n+1}}{n+1} + C$$

$$L = \frac{4}{9} \cdot \frac{u^{1/2+1}}{(1/2+1)} \Bigg|_{13/4}^{10}$$

$$L = \frac{4}{9} \cdot \frac{u^{3/2}}{3/2} \Bigg|_{13/4}^{10} = \dots$$

4: $x = \frac{1}{3} (y^2 + 2)^{3/2}$ $y=0$ to $y=1$

$$L = \int_0^1 \sqrt{1 + \left(\frac{dx}{dy}\right)^2} \cdot dy$$

$$\frac{dx}{dy} = \left[\frac{1}{3} (y^2 + 2)^{3/2} \right]'$$

$$= \frac{1}{3} \cdot \frac{3}{2} (y^2 + 2)^{3/2 - 1} \cdot (y^2 + 2)'$$

$$= \frac{1}{2} (y^2 + 2)^{1/2} \cdot 2y$$

$$= (y^2 + 2)^{1/2} \cdot y$$

$$L = \int_0^1 \sqrt{1 + \left[(y^2 + 2)^{1/2} \cdot y \right]^2} dy$$

$$L = \int_0^1 \sqrt{1 + (y^2 + 2) \cdot y^2} dy$$

$$L = \int_0^1 \sqrt{1 + 2y^2 + y^4} \cdot dy$$

$$L = \int_0^1 \sqrt{(1 + y^2)^2} dy \quad \left[(1 + y^2)^2 = 1 + 2y^2 + y^4 \right]$$

$$L = \int_0^1 (1+t^2) dt$$

$$L = \left[t + \frac{t^3}{3} \right]_0^1 = \frac{4}{3}$$

8 $x = \frac{1}{8} t^4 + \frac{1}{4} t^{-2}$ $t=1$ to $t=4$

$$\frac{dx}{dt} = \left(\frac{1}{8} t^4 + \frac{1}{4} t^{-2} \right)'$$

$$\frac{dx}{dt} = \frac{1}{8} \cdot 4 t^3 + \frac{1}{4} \cdot (-2) t^{-3}$$

$$= \frac{1}{2} t^3 - \frac{1}{2} t^{-3}$$

$$\Rightarrow L = \int_1^4 \sqrt{1 + \underbrace{\left(\frac{1}{2} t^3 - \frac{1}{2} t^{-3} \right)^2}_A} dt$$

$$A = 1 + \left(\frac{1}{2} t^3 - \frac{1}{2} t^{-3} \right)^2$$

$$(a+b)^2 = a^2 + 2ab + b^2$$

$$= 1 + \left(\frac{1}{2} t^3 \right)^2 - \cancel{2} \cdot \cancel{\frac{1}{2}} \cancel{t^3} \cdot \cancel{\frac{1}{2}} \cancel{t^{-3}} + \left(\frac{1}{2} t^{-3} \right)^2$$

$$= 1 + \left(\frac{1}{2} t^3 \right)^2 - \frac{1}{2} + \left(\frac{1}{2} t^{-3} \right)^2$$

$$= \left(\frac{1}{2} y^3\right)^2 + \frac{1}{2} + \left(\frac{1}{2} y^{-3}\right)^2$$

$$= \left(\frac{1}{2} y^3\right)^2 + \underbrace{2 \cdot \left(\frac{1}{2} y^3\right) \cdot \left(\frac{1}{2} y^{-3}\right)}_{1/2} + \left(\frac{1}{2} y^{-3}\right)^2$$

$$= \left(\frac{1}{2} y^3 + \frac{1}{2} y^{-3}\right)^2$$

$$\Rightarrow L = \int_1^4 \sqrt{A} \, dy$$

$$L = \int_1^4 \sqrt{\left(\frac{1}{2} y^3 + \frac{1}{2} y^{-3}\right)^2} \, dy$$

$$L = \int_1^4 \left(\frac{1}{2} y^3 + \frac{1}{2} y^{-3}\right) \, dy$$

$$L = \left. \frac{1}{2} \cdot \frac{y^4}{4} + \frac{1}{2} \cdot \frac{y^{-2}}{-2} \right|_1^4 = \dots$$